

RSC Applied Polymers

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Energy augmentation of triboelectric nanogenerators using PDMS-MWCNT Composites and their applications in IoT and HMI sensing

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Abstract

Triboelectric nanogenerators (TENGs) have become a viable self-powered system which may satisfy the demand of the increasing need for portable and adaptable power sources. By this technique mechanical vibration can be efficiently captured, from locomotive motion of vehicles, human beings, rain falls, ocean waves, air flows etc. Depending on the triboelectric series various materials have been used and explored for the TENG application. In this work, we have investigated the triboelectric characteristics of spin-coated polydimethylsiloxane (PDMS) and multiwall carbon nanotubes (MWCNTs) composite membrane. By adding various concentration of MWCNT in PDMS the charge transfer efficiency was investigated in terms of current output. At the optimized composition of 0.05 wt. % MWCNT in PDMS an open-circuit voltage (V_{OC}) of 110 V, and a short-circuit current (I_{SC}) of 10 μ A were observed which lead to nearly 1 W/m² power density. Additionally, this composition demonstrated outstanding long-term durability and electrical stability which make them appropriate for wearable clothing and shoes which would enable harvesting energies during routine activities jogging or walking as demonstrated.

Keywords—Triboelectric Nanogenerator, Self-powered Sensor, Wearable Sensor, Energy Harvesting, Human Motion Tracking.

Introduction

As an alternate way of powering portable gadgets, healthcare equipment, and wireless sensor systems, energy harvesting technologies have received significant attention in recent years^{1,2}. An extensive study has been devoted in the field of wearable devices to harness energy from human motion³⁻⁶. However, recent years have seen a surge in research into the use of triboelectric nanogenerators (TENG) to convert the motions of the human body into electric energy and use in the power portable devices, internet of things (IoT) gadgets and water proof textile⁷. As we have seen TENGs are flexible, efficient, and cost-effective energy harvesters⁸.



9. One of the most significant obstacles that must be overcome to produce flexible energy harvesting devices is the selection of suitable materials that provide both flexibility and mechanical stability¹⁰⁻¹². Additionally, the design and construction of the whole structure must be accomplished. Significant efforts have been made for the last decade to improve the performance of flexible TENG with the intention of broadening the variety of applications that these nanogenerators may be used for. Polydimethylsiloxane (PDMS) is known as one of the most resilient organic non-metallic flexible materials, making it extremely appropriate as a triboelectric material which intrinsically comes under negative class. It features great flexibility, stretchability, and biocompatibility, which are necessary for wearable energy harvesters^{12, 13}. Furthermore, PDMS possesses very high dielectric constant characteristics, which allow it to simultaneously absorb and store the electrostatic charges that are formed during the triboelectric process. This generation of the charge because of their difference in electronegativity is referred to as the triboelectric process. It has been shown in several studies that the energy conversion capabilities of these devices have been enhanced by surface modification through the process of nano-structuring, which involves increasing the surface area of the parent matrix of TENGs. There have been many research that have demonstrated an increase in the energy conversion efficiency of a PDMS-based TENG using several types of conductive materials^{14, 15}. Multiwall carbon nanotubes, also known as MWCNT, are among one of the most effective materials for improving the electrical characteristics of materials. The atomic structure of MWCNT is one of a kind, and as a result, it boasts outstanding mechanical strength, electrochemical stability, electrical conductivity, and thermal resistance¹⁶⁻¹⁸. As a result of the incorporation of MWCNTs into PDMS, the surface charge density and ability to charge transfer can be significantly boosted, which ultimately results in enhanced performance across a variety of applications. Within the context of PDMS-based materials, this combination makes use of the remarkable properties of MWCNTs to improve the overall functioning and longevity of products.

Multiwall carbon nanotubes (MWCNTs) have emerged as potential conductive materials for improving the performance output of triboelectric nanogenerators (TENGs). Recently, Salauddin et al. proposed a triboelectric nanogenerator (TENG) utilizing MXene-integrated eco-flex composites encased in fabric, with MXene functioning as the tribo-negative layer. The device attained a maximum output power of 3.69 mW at an input impedance of 4.5 MΩ, with an applied force of 8 N and a frequency of 4.5 Hz¹⁹. In a similar way, Yang et al. engineered a flexible hybrid nanogenerator that integrates piezoelectric and triboelectric elements utilizing an Al/PVDF/Cu-PDMS/ITO/PET configuration, achieving a peak current of 20 μA, a voltage of 88 V, and a power output of 98 μW²⁰. Zhu et al. examined PDMS-Ag nanocomposites, yielding an output voltage of 33.6 V, a current of 4.5 μA, and a power density of 162 mW/m² at a pressure of 53.57 kPa²¹. In a separate investigation, Hajra et al. presented a ZIF-67-based triboelectric nanogenerator (TENG) intended for autonomous robotic object identification, attaining a peak output voltage of 118 V, a short-circuit current of 1.7 μA, and a power density of 0.15 W/m²².

This work's innovation is the creation of a high-performance, flexible triboelectric nanogenerator (TENG) utilizing a PDMS-MWCNT composite membrane, optimized by meticulous regulation of MWCNT composition. The incorporation of 0.050 wt.% MWCNT



into the PDMS matrix significantly enhances the triboelectric performance of the TENG, resulting in a power density of 0.97 W/m^2 , an output voltage of 110 V , and a current of $10 \mu\text{A}$, while demonstrating exceptional durability over 10,000 operating cycles. The effective incorporation of the TENG into wearable devices for energy harvesting from human motion underscores its practical use in self-powered sensing technologies. This scalable manufacturing approach distinguishes itself from prior research by providing a dependable, robust, and efficient solution for wearable energy harvesting. Additionally, to replicate authentic motion events, we created and evaluated a wearable prototype.

2. Results and Discussion

2.1. Optimization and Characterization of TENG

A. Structural phase and morphological analysis of the fillers

The structural characterization of the synthesized PDMS-MWCNT composite membrane was carried out by XRD, as shown in Figure 1(a). The XRD pattern of the PDMS material exhibits a peak at a 2θ angle of 11.80° ^{23, 24}. Upon incorporating the MWCNT into PDMS, the XRD pattern of the MWCNT phase aligns with the JCPDS Card No 54-0964. A distinct and well-defined peak corresponding to the C layer is seen in the XRD pattern of MWCNTs at a 2θ angle of 26° , aligned along the preferred direction of (002)²⁵. Additionally, the XRD peaks of PDMS decreased in intensity after incorporating MWCNTs into the composite membrane. The chemical bond and structure of the composite membrane were investigated using FTIR. When the sample was exposed to infrared light, it showed clear peaks, indicating the presence of several chemical connections. Figure 1(b) exhibits a prominent peak at 784.08 cm^{-1} , indicating the oscillatory movement of $\text{Si}(\text{CH}_3)_2$. The existence of Si-C bonds in the PDMS-MWCNT composite membrane was confirmed by the peak at 843.8 cm^{-1} . The detection of Si-O-Si stretching in PDMS was identified by a complex signal seen at 1004.2 cm^{-1} . The occurrence of a peak at 1256.08 cm^{-1} indicates the existence of a vibration related to the distortion of the $-\text{CH}_3$ group in PDMS. Moreover, a conspicuous peak at 2962.02 cm^{-1} was observed, suggesting the presence of MWCNT. The Raman spectra of PDMS and the composite membrane were examined by using a 532 nm laser source with a power output of 0.28 mW , as seen in Figure 1(c) and zoomed view Figure 1(d). The Raman spectrum of the pure PDMS sample exhibited distinct peaks at specific wavenumbers: 491.56 cm^{-1} for Si-O-Si symmetric stretching, 615.48 cm^{-1} for Si-C symmetric stretching, 709.85 cm^{-1} for Si- CH_3 symmetric rocking, 1258.92 cm^{-1} for CH_3 symmetric bending, 1414.57 cm^{-1} for CH_3 asymmetric bending, 2901.85 cm^{-1} for CH_3 symmetric stretching, and 2966.21 cm^{-1} for CH_3 asymmetric stretching²⁶. Upon the addition of MWCNT to the PDMS matrix, similar peaks were seen, but with reduced intensity compared to pure PDMS. Furthermore, an additional peak at 1584.32 cm^{-1} was observed, presumably attributed to the presence of sp^2 hybridized C=C bonds or the elongation of aromatic rings²⁷.



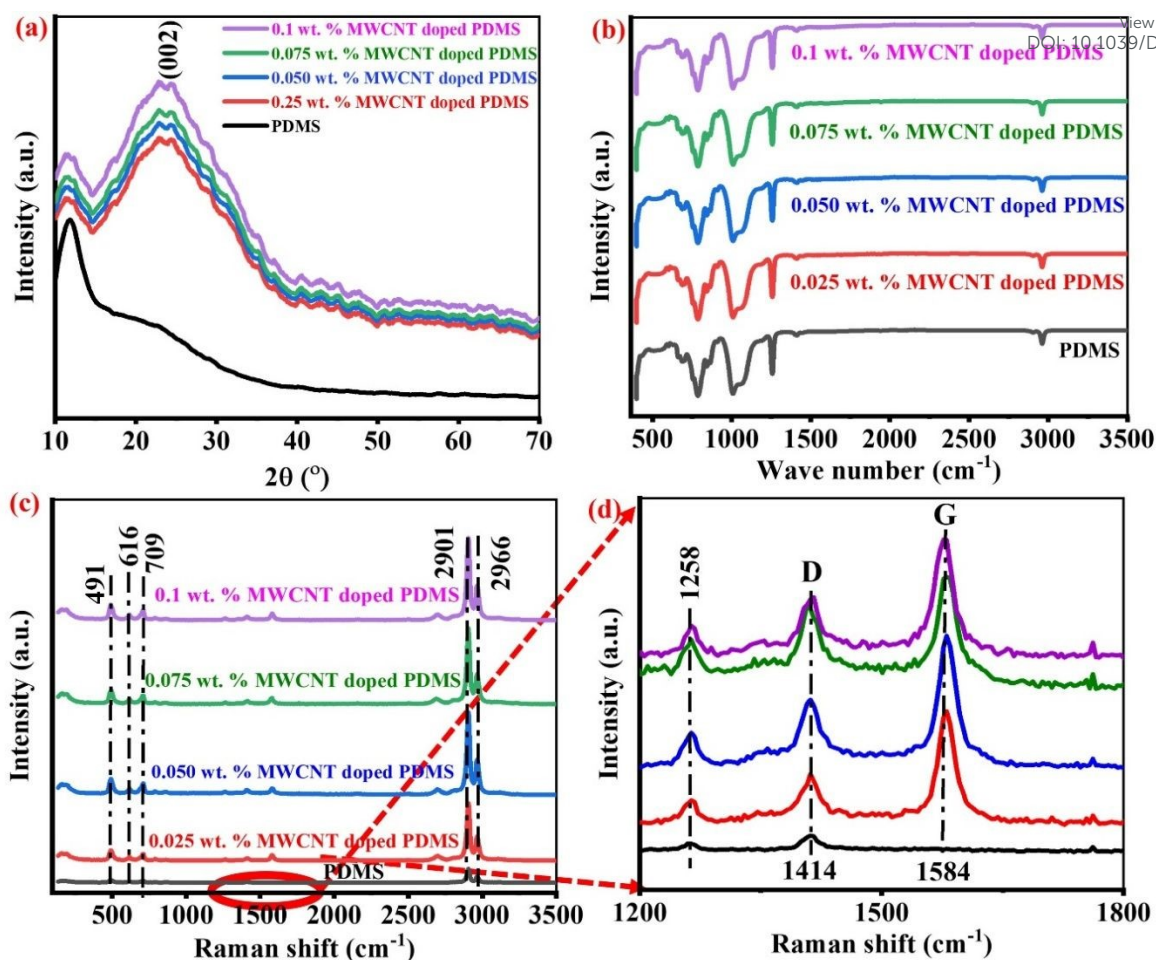


Figure 1: Material characterization of a spin-coated PDMS-MWCNT composite membrane (a) X-ray diffraction analysis, (b) FTIR, (c) Raman spectra, and (d) enlarged view of raman spectra from 1200 to 1800 cm^{-1} .

The morphology of the PDMS-MWCNT composite membranes was investigated by SEM imaging, as shown in Figure 2. The SEM micrographs show a homogeneous dispersion of MWCNT nanofillers inside the PDMS matrix, indicating that the nanotubes were successfully incorporated into the polymer framework.

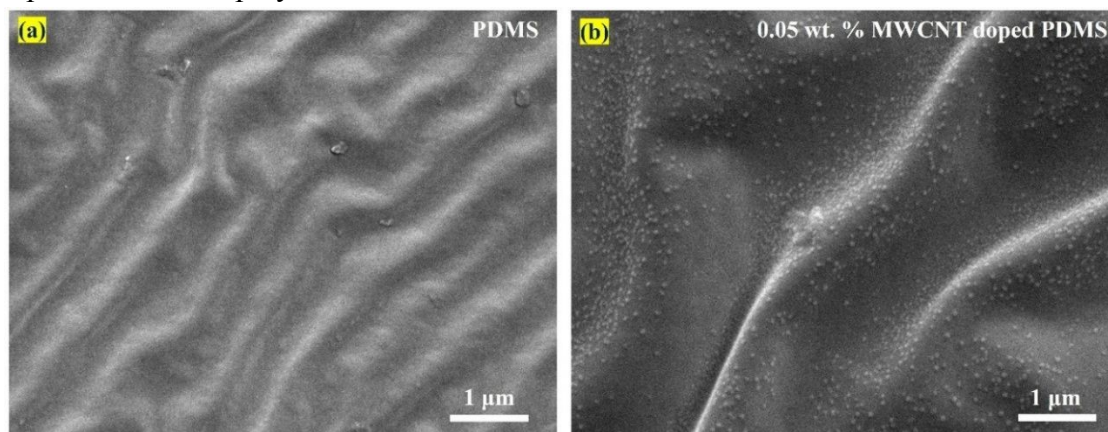


Figure 2: SEM image of (a) Pristine PDMS, and (b) 0.05 wt.% MWCNT doped PDMS.

B. Triboelectric Output Performances



The effective area of the contact surfaces and the choice of contact materials are known to significantly affect the output characteristics of a TENG. To evaluate the triboelectric performance, an in-house triboelectric test setup was used, specifically designed for operation in contact separation mode²⁸. The triboelectric output performance is determined by contact electrification and electrostatic induction. A theoretical model of contact separation mode TENGs is illustrated in Figure 3. This model can be conceptualized as two films with opposing triboelectric properties, each having electrodes attached to their back sides.

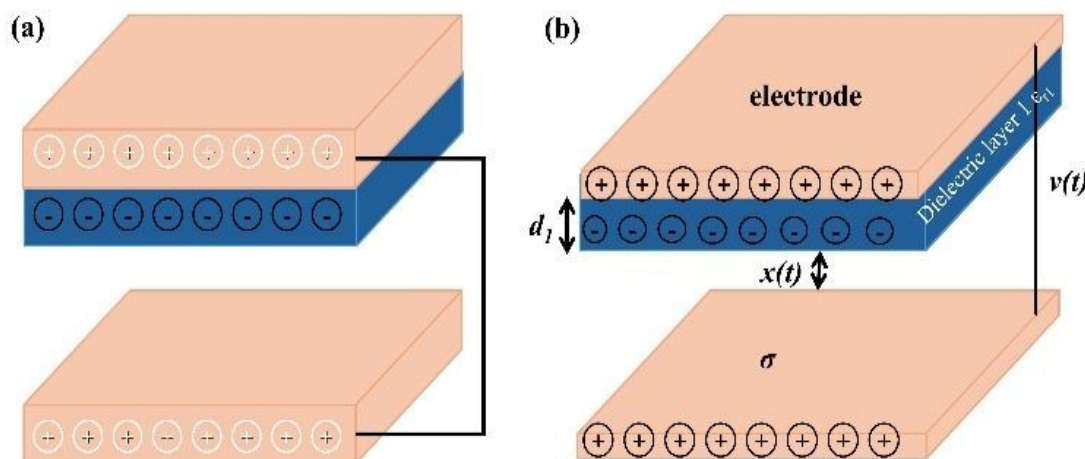


Figure 3: Contact separation mode TENG: (a) structural design of the device, and (b) theoretical model.

The output open circuit voltage and short circuit current are calculated using the parallel capacitor model, represented by the equation V-Q-x^{29,30}.

$$V_{output} = -\frac{Q}{S\epsilon_0} \left(\frac{d_1}{\epsilon_{r1}} + x(t) \right) + \frac{\sigma_T x(t)}{\epsilon_0} \quad (1)$$

In open circuit conditions $Q = 0$, then

$$V_{OC} = \frac{S\sigma_T}{C_{air}} = \frac{\sigma_T x(t)}{\epsilon_0} \quad (2)$$

In the short circuit condition when the output voltage is zero

$$Q_{SC} = \frac{S\sigma_T}{C_{air}C_{total}} = \frac{S\sigma_T x(t)}{d_0 + x(t)} \quad (3)$$

$$I_{SC} = \frac{dQ_{SC}}{dt} = \frac{S\sigma_T d_0 v(t)}{(d_0 + x(t))} \cdot \frac{1}{(d_0 + x(t))} \quad (4)$$

where Q is the transferred free charge, S contact area, d_1 tribo layer thickness, ϵ_{r1} the relative permittivity of tribo layers, ϵ_0 air permittivity, $x(t)$ separation distance, $v(t)$ vibrational speed, and σ_T tribo charge density.

The electrical performance of the TENGs has been evaluated by measuring the open-circuit voltage (V_{OC}) and short-circuit current (I_{SC}) at a frequency of 4.4 Hz, as seen in Figure 4. The triboelectric output was measured using metal electrodes (copper and aluminium) in contact with the MWCNT-doped PDMS composite membrane. As shown in Supplementary Table S1 and Figure S1, the output performance of copper and aluminium was comparable. Thus, copper



was chosen as the electrode material for all future triboelectric performance testing. The top electrode and friction layer had a contact surface of $20 \times 20 \text{ mm}^2$, separated by 6 mm. The output performance was investigated at different wt. % of MWCNT (0.025 wt. %, 0.050 wt. %, 0.075 wt. %, and 0.10 wt. %) into PDMS matrix. The corresponding open circuit voltage (Figure 4(a-e)) and short circuit current (Figure 4(f-j)) values were recorded as follows: 54 V, 84 V, 110 V, 70 V, and 62 V, and $5.0 \mu\text{A}$, $5.8 \mu\text{A}$, $10.0 \mu\text{A}$, $6.6 \mu\text{A}$, and $5.8 \mu\text{A}$, respectively. The highest V_{OC} ($\sim 110 \text{ V}$) and I_{SC} ($\sim 10.0 \mu\text{A}$) were achieved at 0.050 wt. % MWCNT, attributed to the optimized distribution of MWCNTs within the PDMS matrix at this concentration which facilitate charge transfer efficiently. The decrease in V_{OC} and I_{SC} were observed wt. % greater than 0.075 wt. % due to exceeding in percolation threshold. Beyond the percolation limit, the composite membranes start losing their properties³¹. Notably, the TENG with 0.050 wt. % MWCNT exhibited nearly double the output voltage and current compared to the TENG fabricated solely with PDMS.

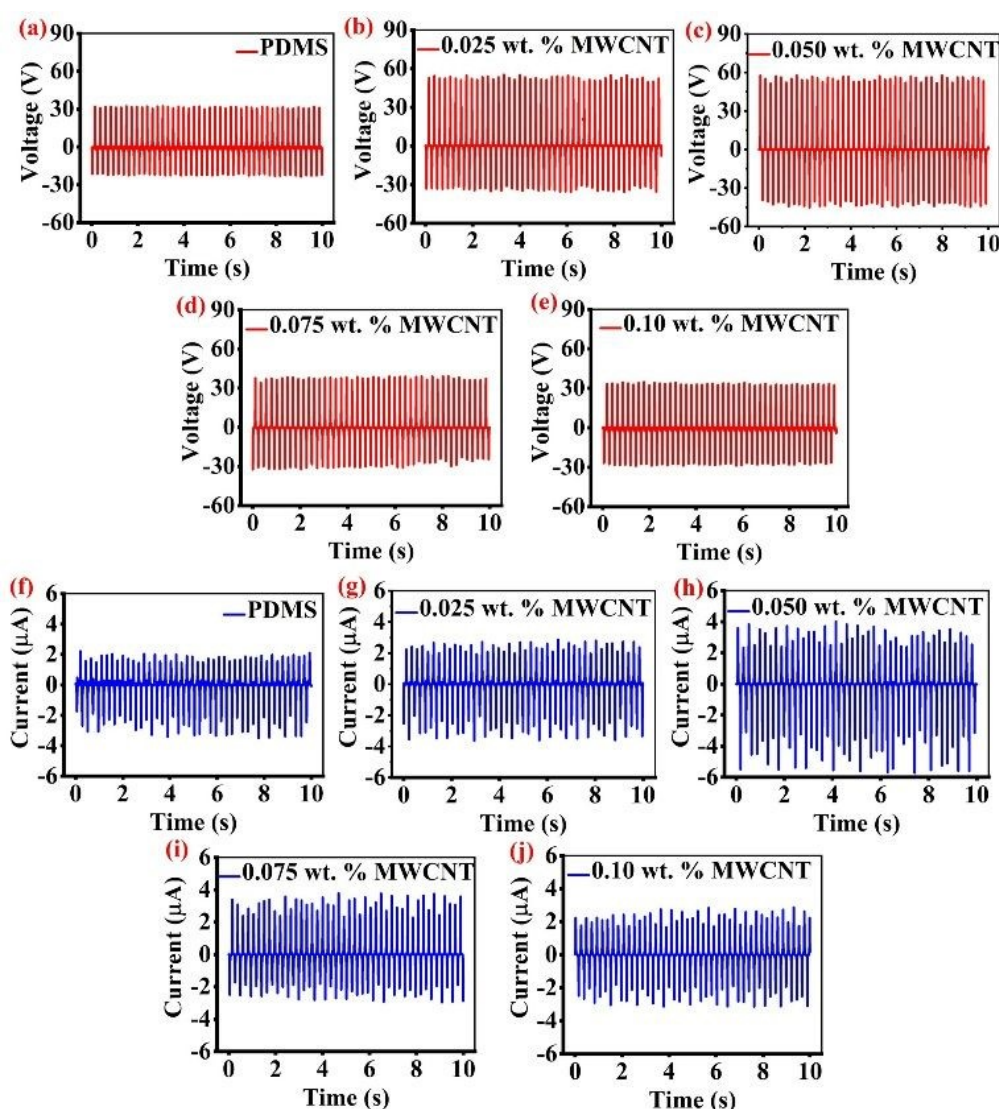


Figure 4: Open circuit voltage (V_{OC}) and short circuit current (I_{SC}) of different weight percentages of MWCNT doped PDMS.



We further investigated the output performance of 0.050 wt.% MWCNT-doped PDMS at varying distances between the two triboelectric layers, ranging from 2 mm to 10 mm, as illustrated in Figure 5. The highest V_{OC} and I_{SC} outputs were observed at a separation distance of 6 mm, as depicted in Figures 5(a) and 5(d). Additionally, we studied the effect of different external pressures, ranging from 1 N to 20 N. The voltage output reached a maximum of 110 V, while the current peaked at 10 μ A at an applied force of 15 N, as shown in Figures 5(b) and 5(e). Furthermore, we optimized the rotational velocity of the linear motor at four speeds: 20.93 rad/s, 41.87 rad/s, 62.8 rad/s, and 83.73 rad/s. The maximum V_{OC} and I_{SC} were achieved at a rotational velocity of 62.8 rad/s, as illustrated in Figures 5(c) and 5(f). Over a frequency range of 1.8 to 4.4 Hz, the open-circuit voltage (V_{OC}) and short-circuit current (I_{SC}) were recorded in the same way. V_{OC} of 55 V and I_{SC} of 5.5 μ A were seen in the TENG at lower frequencies. At higher frequencies, they went up to 110 V and 10.0 μ A. The related data and graphs can be found in the Supplementary Material, which has Supplementary Table S2, Figure S2, and Figure S3.

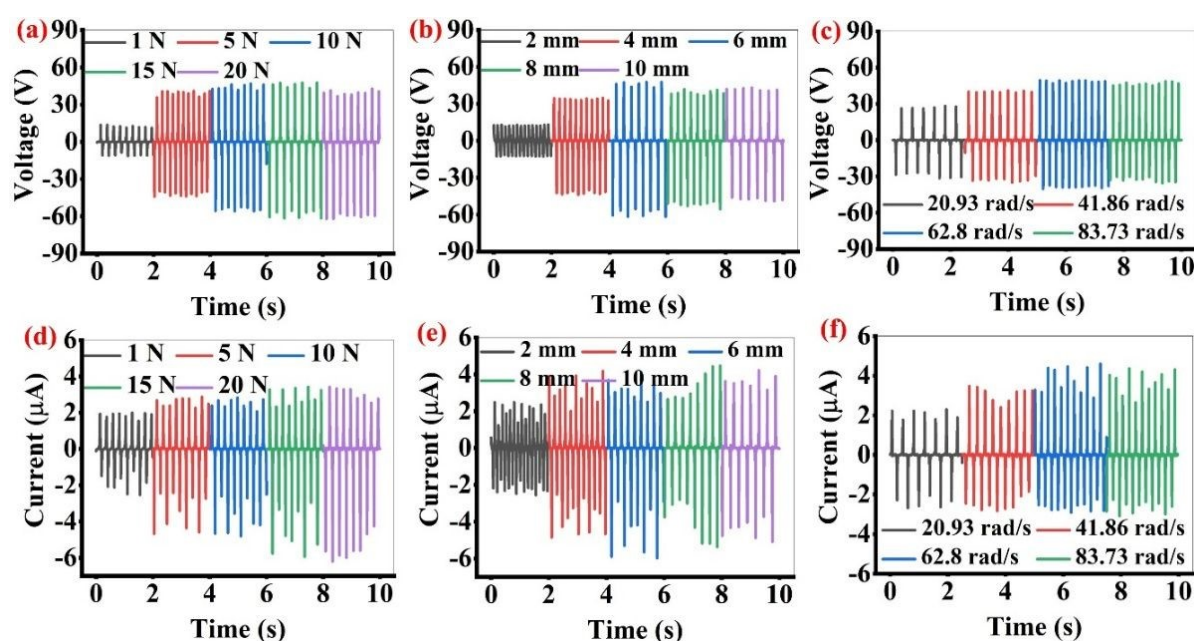


Figure 5: V_{OC} and I_{SC} at different separating distances, different applied forces, and different angular velocities.

To assess the energy harvesting capability of the CS-TENG, the relationship between output voltage (V) and load resistance (R) was measured, and the power density (P.D) was calculated using the formula $P.D = V^2/R.A$, where A is the device area, as depicted in Figure 6(a). The P.D was determined at a 6 mm separation between the triboelectric layers and an applied frequency of 4.4 Hz under varying external loads. The output voltage exhibited a steady increase with increasing R across the entire range. Correspondingly, the power density initially rose with R, reaching peak values of 0.2 W/m² for pure PDMS and 0.98 W/m² for PDMS containing 0.050 wt.% MWCNT, both at an optimal resistance of 5 M Ω . The impedance of the device was further investigated by analyzing the relationship between voltage, current, and external load, yielding a value of 5 M Ω , as shown in Figure 6(b). The aging effect on TENG samples was investigated by measuring the output performance parameters weekly for 10



weeks as illustrated in Figure 6(c). A gradual decrease of up to 4 % in output was observed after seven weeks. The maximum duration of 20 % in V_{OC} was noticed after ten weeks. The device output voltage was recorded across various humidity levels (ranging from 20 % to 70 %) and over different time intervals (up to 10 weeks) to assess its long-term stability as shown in Figure 6(d). Notably, the TENG generated a voltage of 110 V at 20 % humidity, but the device's response significantly declined with increasing humidity. At 70 % humidity, the TENG's voltage dropped to 95 V. Higher relative humidity exacerbates the dissipation of triboelectric charges on the friction layers, leading to a general decrease in the output of the TENG. Further, we examined the impact of external forces and constant frequency on its output performance. Additionally, a durability test involving 10,000 cycles was conducted, and the results are presented in Figure 6(e). Remarkably, there was no degradation in the output response over the entire 10,000 cycles, affirming the robustness of the fabricated TENG. This stability is particularly evident when operating at significantly lower frequencies, with minimal influence of frequency variations. The obtained output parameters were compared with the existing available PDMS -based TENGs in Table 1.

Table 1: Performance parameters (area, voltage, current, impedance, and average power density) of selected examples of triboelectric materials based on PDMS composite membranes.

Material	Area (cm ²)	Voltage (V)	Current (μA)	Impedance (MΩ)	Average Power Density (W/m ²)	Ref.
PDMS/BaTiO ₃	100	2500	150	5	1.2	32
PDMS/CNT	6	77.8	25.7	3	3.3	12
PDMS/SrBaTiO ₃	90	280	8.5	100	4.4	33
PVDF/3D ZnO	6	17	2.5	-	-	34
Teflon-PDMS/ZIF-67	-	118	1.7	50	0.15	22
Al/PVDF/Cu- PDMS/ITO/PET	-	88	20	4.5	-	20
PDMS/AGs nanowires	20	33.6	4.5	8.6	0.23	21
PDMS/MWCNT	4	110	10	5	0.98	This work



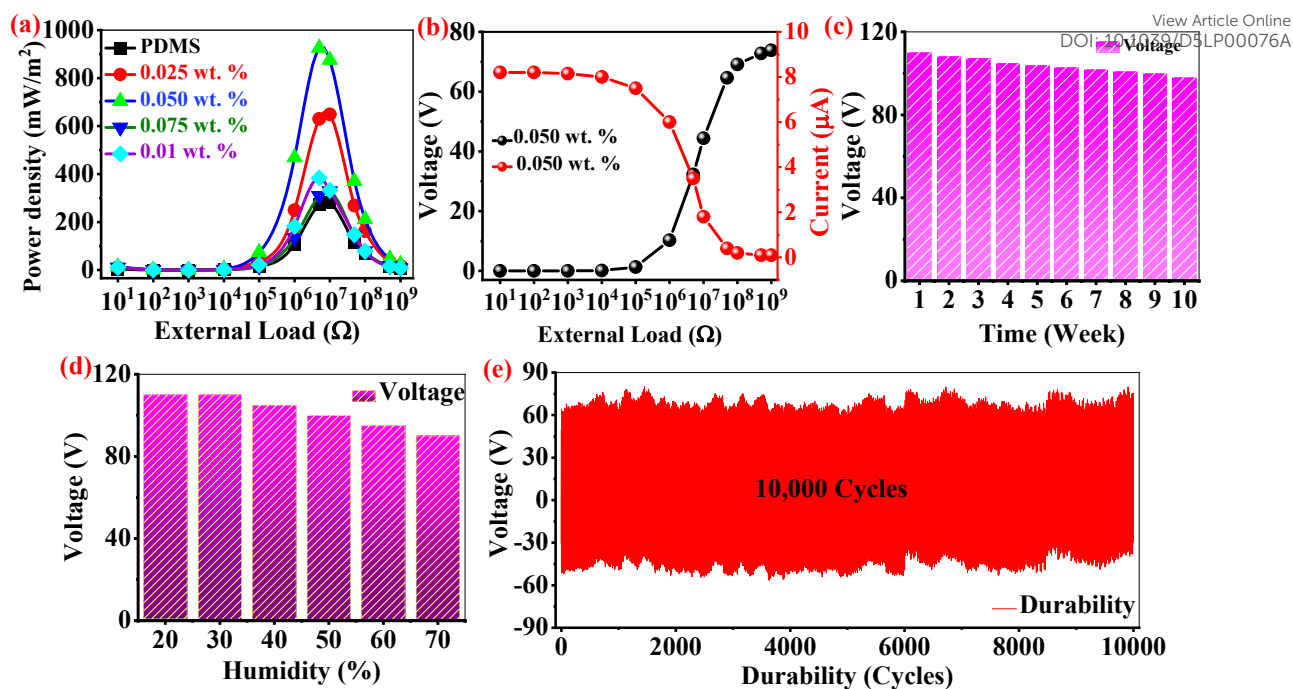


Figure 6: (a) Power density at different load resistance for all composite membranes, (b) impedance of composite PDMS-MWCNT (0.050 wt. %), (c) Durability test of PDMS-MWCNT (0.050 wt. %) for 10 weeks (d) Effect of Humidity on the output voltage, and (e) Durability test for 10,000 cycles.

Furthermore, the accumulated surface charge of pure PDMS and 0.050 wt. % MWCNT is calculated by using the following equation³⁵.

The accumulated charge during one half of the AC cycle: Integrate over half a cycle $t = 0$ to $t = T/2$

$$Q_{half} = \int_0^{T/2} I_{max} \sin(\omega t) dt \quad (5)$$

Substitute $\omega = 2\pi f$

$$Q_{half} = \frac{I_{max}}{2\pi f} [-\cos(2\pi f t)]_0^{T/2} \quad (6)$$

$$Q_{half} = \frac{I_{max}}{2\pi f} [-(-1 + 1)] = \frac{2I_{max}}{2\pi f} = \frac{I_{max}}{\pi f} \quad (7)$$

For PDMS

$$Q_{full} = \frac{2I_{max} \times T}{\pi} = \frac{2 \times 4.5 \times 10^{-6} \times 0.02}{3.14} = 0.05 \mu C$$

For 0.050 wt. % MWCNT

$$Q_{full} = \frac{2I_{max} \times T}{\pi} = \frac{2 \times 10 \times 10^{-6} \times 0.04}{3.14} = 0.25 \mu C$$

The 0.050 wt. % MWCNT has shown the maximum accumulated charge and charge density i.e. $0.25 \mu C$ and $637 \mu C/m^2$ in one complete alternating cycle. The overall performance of the proposed device is assessed by calculating the figure of merit (FOM). This parameter, influenced by the surface properties of the materials, is evaluated for the device (FOM_{Device}) and the material ($FOM_{Material}$). The calculations are carried out for pure PDMS and PDMS incorporating 0.05 wt. % MWCNT using the given expression (8-13).



$$FOM_{Device} = 0.064 \times \frac{\sigma_T^2 V_m}{\epsilon_0} \quad (8)$$

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DOI: 10.1039/D5LP00076A

$$FOM_{Material} = \frac{\sigma_T^2}{\epsilon_0} \quad (9)$$

$$\eta = \frac{E_{output}}{E_{input}} \quad (10)$$

$$E_{output} = Power \times Time \quad (11)$$

$$E_{input} = force \times gap \text{ between the two plates, } (12)$$

$$V_m = \omega_r \cdot \frac{X_{max}}{\pi} \quad (13)$$

Here, ϵ_0 represents the permittivity of vacuum, σ_T is the surface charge density, V_m is the velocity of the contact separation impact motion. In this expression, $X_{max} = 6 \text{ mm}$, is the maximum gap between the two materials, and $\omega_r = 2\pi f^{36}$. The FOM_{Device} and $FOM_{Material}$ values of the proposed pure PDMS and 0.05 wt. % MWCNT are 0.97 W/m^2 , 1.7 kJ/m^3 , and 70 W/m^2 , 45 kJ/m^3 and efficiency 1.0 % respectively.

3. BIOMECHANICAL ENERGY HARVESTING APPLICATIONS

Human body motion, typically characterized by low frequencies, can be effectively harvested by the CS-TENG and stored for later use. To illustrate this capability, common forms of biomechanical energy such as hand tapping, walking, and running were employed as examples. The circuit configuration is depicted in Figure 7(a), where the output of the CS-TENG was connected to one terminal of a full-wave rectifier circuit, while the other terminal was left floating. The rectified output was then utilized to operate a stopwatch powered by the TENG. In Figure 7(b), the rectified output was also connected to various capacitors, ranging from 1.0 to 10.0 μF , for charging purposes. These capacitors can subsequently power electronic devices. A 1 μF capacitor was charged to 4.2 V within 120 s, causing a group of LEDs to illuminate, as shown in Figure and Video S1. Furthermore, the TENG device was integrated into a shoe and tested under different human motion conditions, such as walking, jogging, and running. The device generated output voltages of 5 V, 10 V, and 20 V, respectively, for these activities as shown in Figure 6(c and d).



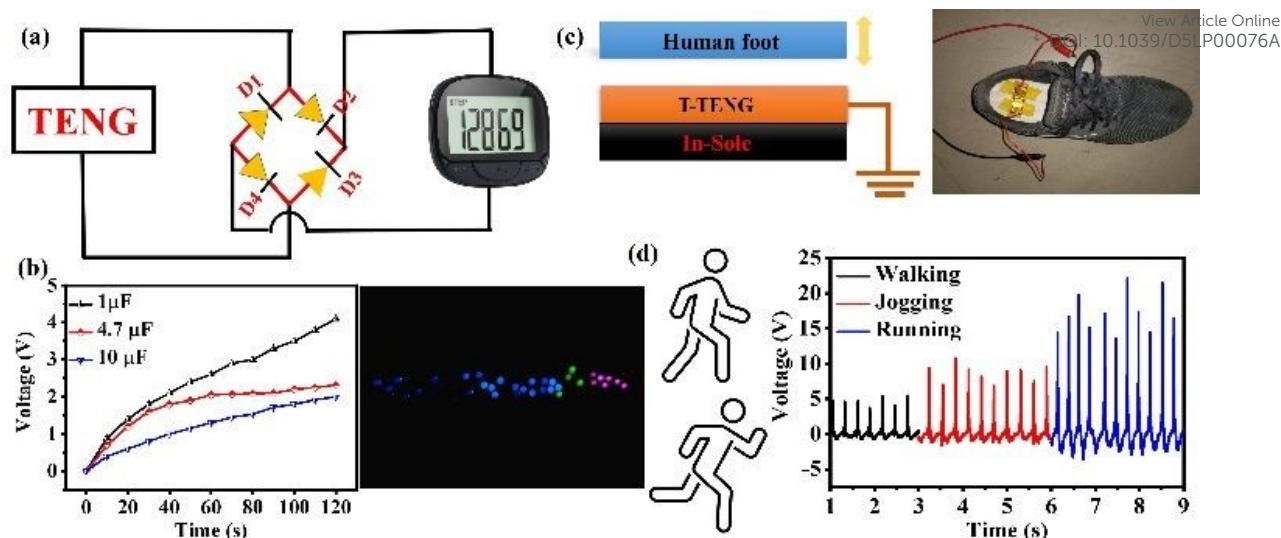


Figure 7: Application of the TENG device: (a) Schematic illustration of a TENG used for charging a stopwatch, (b) Demonstration of capacitor charging using the TENG, (c) Architectural design of a TENG integrated into a shoe for wearable applications, and (d) Output voltage performance under different motion conditions, including walking, jogging, and running.

The TENG device was employed to trigger an action—specifically, toggling an LED as shown in Figure 8(a). When subjected to an applied force, the TENG device generated voltage spikes, each comprising a positive voltage peak followed by a negative one. The amplitude of these spikes varied depending on the magnitude of the applied force. To ensure compatibility with the microcontroller's logic level, the voltage spikes were processed through a full-wave rectifier, converting all negative voltage peaks into positive ones to produce a unidirectional voltage signal as illustrated in Figures 8(b) and (d).

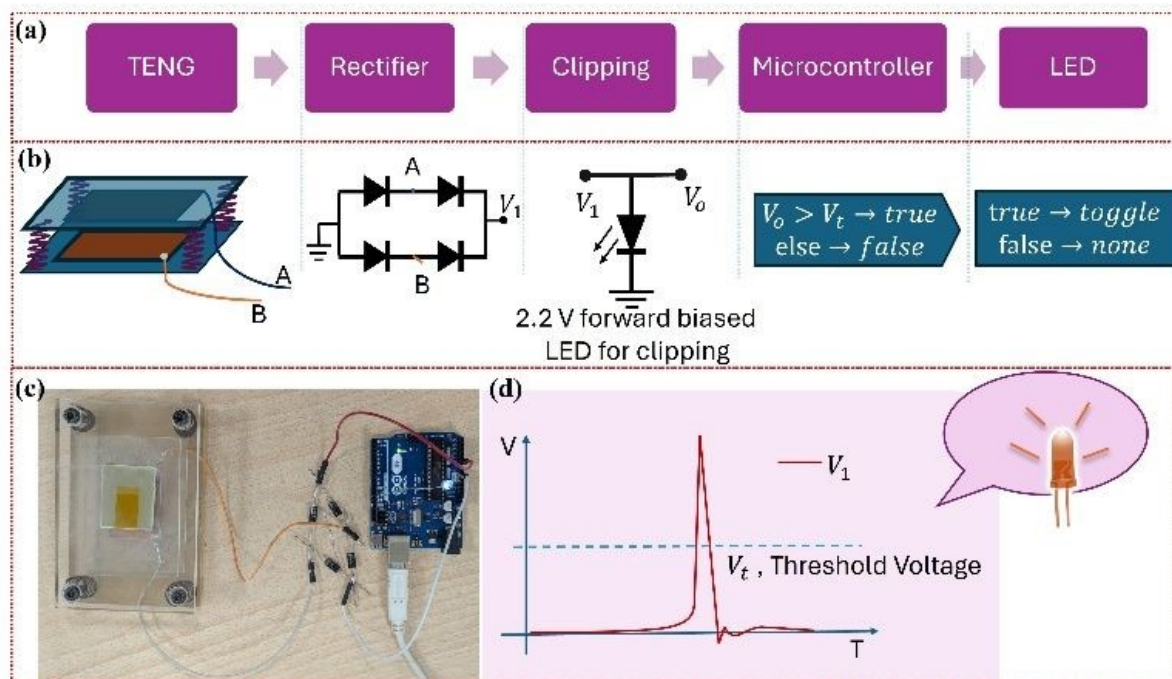


Figure 8: Applications (a) Action trigger mechanism illustrated with an LED response, (b) Representation of a TENG device along with its circuit schematic, (c) Visual depiction of the



sensing device setup, and (d) Demonstration of the response mechanism based on the applied logic.

A forward-biased LED was then used to clip the maximum voltage to 2.2 V, protecting the microcontroller from excessive voltage levels. The microcontroller continuously monitored the rectified and clipped voltage via its analog-to-digital converter (ADC). When the input voltage exceeded the threshold of 2.0 V, the microcontroller toggled the state of an external LED connected through a current-limiting resistor. An LED can be controlled by an HMI system to switch on and off³⁷. This configuration enabled reliable detection and response to the force-induced voltage spikes generated by the TENG device. A visual depiction of the sensing device setup is shown in Figure 8(c) and Video S2. This demonstration showcased the potential of TENG as a self-powered sensor in the HMI system.

4. Conclusions

To summarise, we developed a composite PDMS-MWCNT-based TENG that has improved triboelectric characteristics after conducting our research. By integrating 0.025, 0.050, 0.075, and 0.10 weight % of MWCNT into the PDMS matrix, several different composite PDMS-MWCNT membranes were created. The composition that included 0.050 wt. % of MWCNT had the most impressive performance among the five TENG samples. It was able to achieve the maximum power density of 0.97 W/m², I_{sc} of 10.0 μ A, and V_{OC} 110 V. It was found that the better charge transfer that was made possible by the MWCNT acting as a conductive filler was responsible for the increased triboelectric output. In addition, the TENG demonstrated remarkable stability, being able to keep its performance consistent across 10,000 cycles. The fabricated TENG was integrated with the human body to harvest output voltage signals generated during various human motions, demonstrating its potential for signal-sensing applications under applied forces.

5. Materials and Methods

A. Synthesis of nanocomposite films

The nanocomposite films were fabricated by combining PDMS (Sylgard 184) acquired from Jain Chemicals with MWCNTs with an outer diameter ranging from 20 to 30 nm, procured from Nano Shel, Inc., India.

B. Synthesis of pure PDMS film

A PDMS film was developed by combining the elastomer and curing agent in a 1:10 weight ratio. After physically combining the components for 10 min, the mixture underwent sonication for 20 min to remove any air bubbles. Subsequently, it was poured into a hollow that had been created on a sheet covered with copper. It was then evenly distributed using a spin coater. The film underwent a thermal treatment on a hot plate at a temperature of 100 °C for 2 hours and then underwent a post-treatment at room temperature for 6 hours.

C. Synthesis of MWCNT-PDMS composite film



The PDMS-MWCNT membrane was fabricated by dispersing MWCNT in a chloroform solution. In the process of creating the PDMS-MWCNT composite, MWCNT were usually dissolved in 10 mL of chloroform at a concentration of 0.5 mg/mL. Following vacuum desiccation, the chloroform was allowed to evaporate at room temperature. Sonication was used for 20 min to volatilize the chloroform. The PDMS base was then mixed with the PDMS curing agent at a weight ratio of 10:1. To get rid of any air bubbles, the PDMS mixture was sonicated for a further 20 min. Following that, it was poured into a hollow made on a copper-coated sheet and let to cure for two hours at 100 °C on a hot plate.

D. Fabrication of CS-TENG

The fabrication process of the CS-TENG is illustrated in Figure 8. Initially, a copper-clad sheet with an area of 16 cm² was cleaned using acetone and ethanol to remove any dust particles from its surface. The cleaned sheet was placed on a spin coater, vented, and rotated at a speed of 200 rpm for 120 sec. A prepared MWCNT-PDMS composite solution was carefully dropped onto the surface of the copper-clad sheet. Once the solution was uniformly dispersed across the surface, the sheet was removed from the spin coater and dried on a hot plate at 80°C for 2 h. After drying, the sample was cut into smaller pieces with an area of 4 cm². For CS-TENG operation, two surfaces are required to facilitate contact electrification. Therefore, two materials were selected: the composite film, serving as the negative triboelectric material, and copper, chosen as the positive material based on the triboelectric series. Copper was selected as the electrode material in the PDMS-MAX phase-based TENG, even though it is slightly less negative than aluminium in the triboelectric series. While aluminium could potentially offer higher triboelectric output due to its greater polarity difference with PDMS, copper presents several advantages that make it more suitable for practical applications³⁸. It offers excellent electrical conductivity, better resistance to oxidation, and enhanced chemical stability, which are critical for maintaining consistent performance over time. Additionally, copper provides stronger adhesion to both PDMS and MAX phase materials, improving the mechanical robustness and durability of the device under repeated use. Further, the oxidation potential of copper and aluminium are -0.34 V and -1.66 V respectively which means aluminium has high tendency of getting oxidized than copper which may not be a good as electrode for longer lifetime. These factors make copper a more reliable and efficient choice for flexible and wearable triboelectric nanogenerators. A conductive layer was established by connecting a jumper wire at the interface at the positive and negative triboelectric layers. Finally, the device was connected to a measuring instrument via two wire terminals to evaluate its output performance.



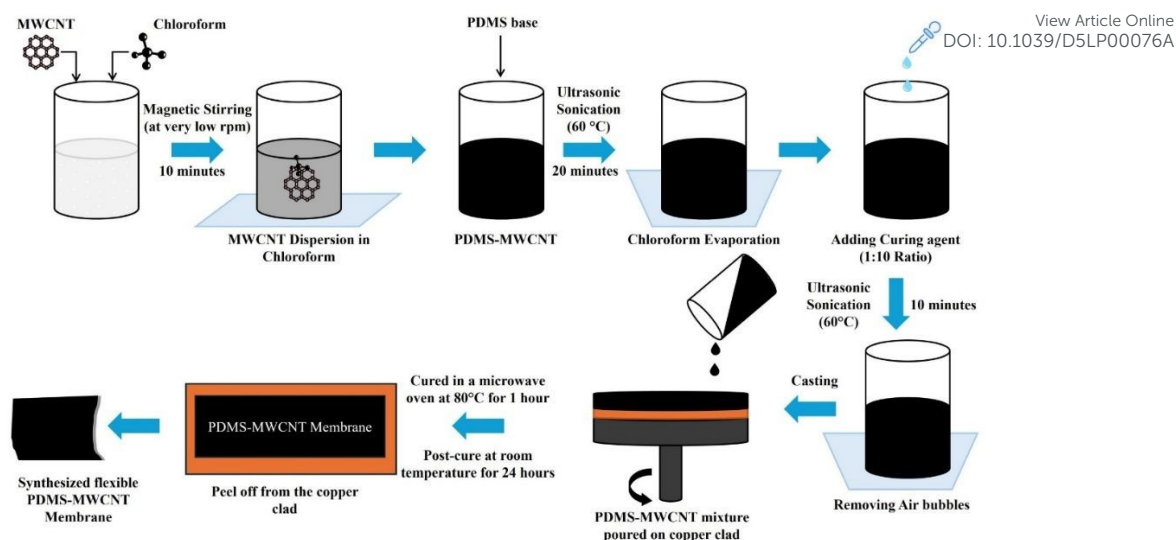


Figure 9: Fabrication procedure of flexible PDMS-MWCNT composite membrane.

E. Working mechanism of TENG

The working mechanism of CS-TENG is illustrated in Figure 9. In State 1, the triboelectric layers are not in touch, resulting in no transmission of charge across the two electrodes. By the triboelectric effect, charge transfer occurs between one layer and another layer in state 2, provided that an external load is given to the system. With the removal of the external load, the triboelectric materials begin to separate from one another, which results in the distribution of the electrical equilibrium. Additionally, as both materials begin to separate from one another, the output voltage tends to grow. By restoring the composite polymer to its original form whenever both components are completely separated from one another, we can achieve the highest possible output voltage, as seen in state 4. When we apply the external load once again, the voltage tends to decrease as shown in state 5. Between the triboelectric layers, this method consistently displays the alternating current (AC) electrical output²⁴.

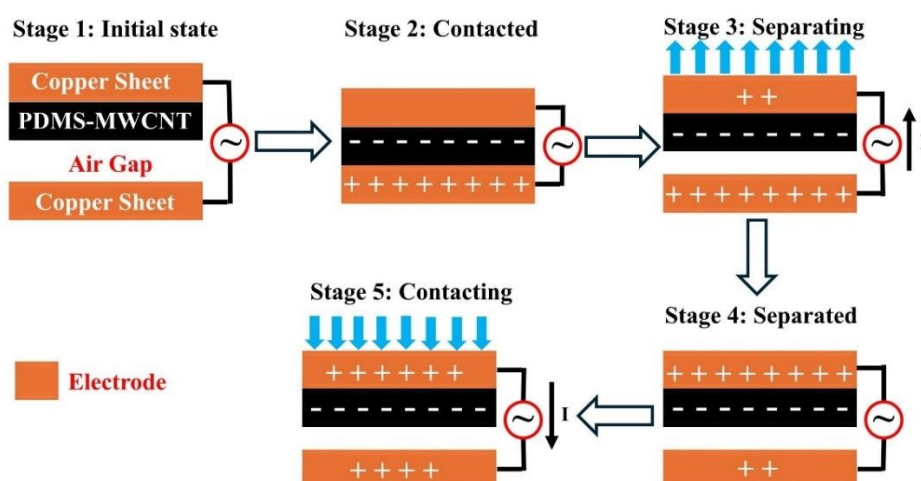


Figure 10: Distributions of transient charge in each tribological layer in a sequential manner, along the operating principle of the device.

F. Characterizations Details



The structural properties of the synthesized materials were determined using an X-ray diffractometer (XRD) method. This technique utilized a Rigaku X-ray diffractometer and Cu α radiation ($\lambda = 1.54 \text{ \AA}$) in the angular range of 10° - 70° at a scanning rate of 3° per min. A Raman spectrum was produced using a Wi Tech alpha 300R confocal Raman Microscope, a laser with a wavelength of 532 nm, an integration time of 10, and a 100x objective. The PDMS membrane and the PDMS-MWCNT composite membrane were examined using Fourier Transform Infrared Spectroscopy (FTIR). The scanning was carried out with the assistance of a VERTEX 80v instrument (Bruker, Germany) that was fitted with a horizontal Attenuated Total Reflectance (ATR) assessor. The scanning was carried out at room temperature and included the wavenumber range of 3500 - 300 cm^{-1} .

Author Declarations

Conflict of Interest

The authors have no conflicts to disclose.

Data Availability

The data that support the findings of this study are available within the article and its supplementary material.

Acknowledgements

The authors express gratitude to the National Research Facility and Central Research Facility at the Indian Institute of Technology Delhi for the XRD measurement. This research was funded by Start-up Research Grant [No. SRG-(2019/001984)]. SERB-DST, CRG (CRG/2023/006778), FIST (SR/FST/ET-I/2020/628), SEED Grant, FIRP, and MFIRP grant from IIT Delhi. The experiment was conducted in the Advanced Electronics Materials and Systems (AEMS) laboratory, Department of Materials Science and Engineering, IIT Delhi.

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Energy augmentation of triboelectric nanogenerators using PDMS-MWCNT Composites and their applications in IoT and HMI sensing

View Article Online

DOI: 10.1039/D5LP00076A

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Data Availability Statement

The data that support the findings of this study are available within the article and its supplementary information.

